

OSLAB PRACTICAL-4 PROGRAMS & OUTPUT

Date: 08-08-26

Program-1 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_4$ cat program1.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int i;

    for(i = 0; i < 3; i++)
    {
        if(fork() == 0)
        {
            printf("Child %d: PID = %d\n", i+1, getpid());
            return 0;
        }
    }

    for(i = 0; i < 3; i++)
        wait(NULL);

    printf("All children completed.\n");

    return 0;
}
sainath@Sainath:~/OSLAB/practical_4$ ./program1
Child 1: PID = 839
Child 2: PID = 840
Child 3: PID = 841
All children completed.
```

Program-2 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_4$ cat program2.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t p[3];
    int i;

    for(i = 0; i < 3; i++)
    {
        p[i] = fork();

        if(p[i] == 0)
        {
            printf("Child %d: PID = %d\n", i+1, getpid());
            return 0;
        }
    }

    for(i = 0; i < 3; i++)
        waitpid(p[i], NULL, 0);

    printf("All children completed.\n");

    return 0;
}
sainath@Sainath:~/OSLAB/practical_4$ ./program2
Child 1: PID = 844
Child 2: PID = 845
Child 3: PID = 846
All children completed.
```

Program-3 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_4$ cat program3.c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    if(fork() == 0)
    {
        printf("Child exiting...\n");
        exit(0);
    }
    else
    {
        printf("Parent sleeping...\n");
        sleep(20);
    }

    return 0;
}
sainath@Sainath:~/OSLAB/practical_4$ ./program3
Parent sleeping...
Child exiting...
```

Program-4 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_4$ cat program4.c
#include <stdio.h>
#include <unistd.h>
#include <stdlib.h>
#include <sys/wait.h>

int main()
{
    if(fork() == 0)
    {
        printf("Child exiting...\n");
        exit(0);
    }
    else
    {
        wait(NULL);
        printf("Child collected.\n");
    }

    return 0;
}
sainath@Sainath:~/OSLAB/practical_4$ ./program4
Child exiting...
Child collected.
```

history:

```
238 cd OSLAB
239 mkdir practical_4
240 cd practical_4
241 gedit program1.c
242 gcc program1.c -o program1
243 ./program1
244 gedit program2.c
245 gcc program2.c -o program2
246 ./program2
247 gedit program3.c
248 gcc program3.c -o program3
249 ./program3
250 gedit program4.c
251 gcc program4.c -o program4
252 ./program4
253 cat program1.c
254 ./program1
255 cat program2.c
256 ./program2
257 cat program3.c
258 ./program3
259 cat program4.c
260 ./program4
261 history
```